EXECUTIVE SUMMARY: Radiomic Feature Sensitivity and Model Interpretability Across Photoacoustic and CT Imaging

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1 Motivation

Radiomics enables quantitative analysis of medical images to characterise tumours non-invasively, but its effectiveness hinges on the stability and interpretability of extracted features. This project aimed to investigate radiomic feature sensitivity and model-driven interpretability across two imaging modalities: photoacoustic imaging (PAI) of patient-derived breast tumour xenografts and clinical CT scans from the LIDC-IDRI lung dataset. In Part A, we reproduced a recent radiomic sensitivity study using full-factorial ANOVA and SHAP-based model interpretation to validate key features in PAI. In Part B, we extended the analysis to a more complex, imbalanced clinical CT dataset, applying multi-stage feature selection, SMOTE class balancing, and SHAP to evaluate feature robustness under realistic conditions. Together, the two parts assess the feasibility of identifying interpretable and generalisable radiomic biomarkers across imaging modalities.

2 Part A: Photoacoustic Imaging Analysis

In Part A, we aimed to reproduce the findings of Escudero Sánchez et al. using photoacoustic images of breast cancer patient-derived xenografts (PDXs). The analysis had two core aims: (1) to assess feature sensitivity using full-factorial ANOVA, identifying features driven primarily by tumour biology rather than confounding variables such as reconstruction method, wavelength, or grey-level binning; and (2) to evaluate discriminative power through SHapley Additive Explanations (SHAP), highlighting features with consistent and meaningful influence on model predictions.

Sensitivity analysis We performed a full-factorial ANOVA on 93 radiomic features extracted from 2268 Volumes of Interest (VOIs) to quantify the proportion of variance (η^2) explained by each factor. As shown in Fig.1, the majority of features were more strongly influenced by acquisition parameters than by tumour model. Grey-level binning emerged as the dominant confounding factor for texture features, often contributing the largest share of variance, while reconstruction method had a particularly strong impact on most first-order statistics.

Despite this, a small subset of features demonstrated high tumour sensitivity with minimal confounding. In particular, the first-order features *Skewness* and *Kurtosis* showed exceptional model sensitivity ($\eta^2 > 0.8$), consistent with findings from Escudero Sánchez et al. Among texture-based features, only *NGTDM Coarseness* exceeded a model sensitivity of 0.6, highlighting it as a rare higher-order metric with consistent tumour relevance in this dataset.

Marginal ANOVA analysis confirmed the trends observed in the full-factorial results. Most higher-order texture features showed notable improvements in tumour model sensitivity when grey-level binning was standardised, particularly at higher binning levels. Conversely, all first-order features showed substantial gains in sensitivity under specific reconstruction methods. These findings indicate that both texture and intensity based features are strongly influenced by image acquisition and processing settings, highlighting the importance of standardisation for reliable radiomic analysis. Importantly, *Skewness* and *Kurtosis* remained consistently stable across all binning levels, while *NGTDM Coarseness* stood out once again as the sole texture feature with reliable performance under both reconstruction variants.

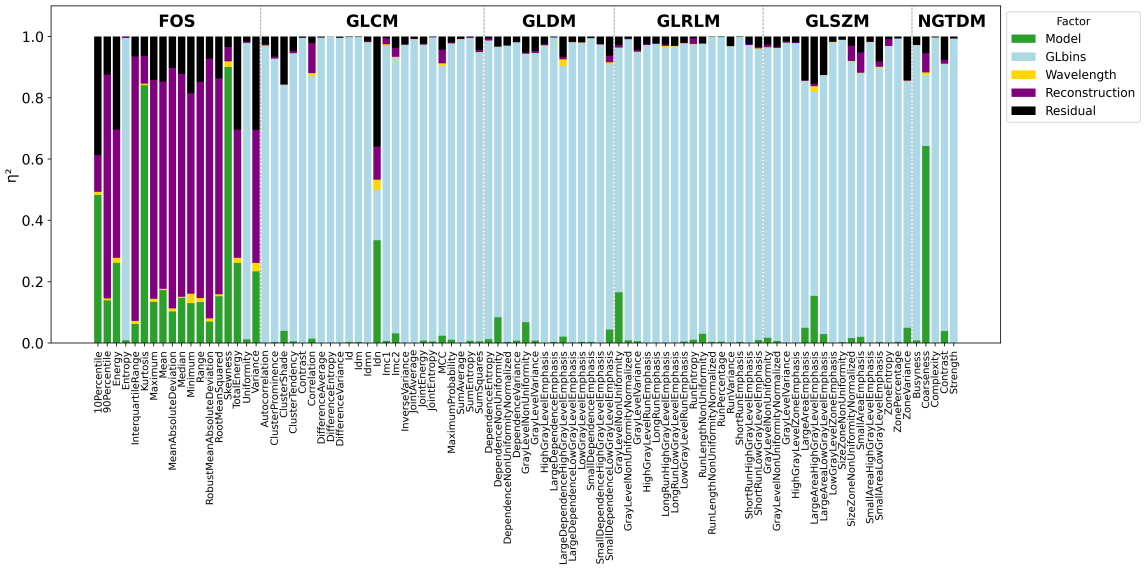

Figure 1: Full factorial ANOVA η^2 contributions for all 93 radiomic features.

SHAP Analysis To complement the ANOVA results, we applied SHAP to Random Forest models trained on the selected features. SHAP confirmed the dominant influence of first-order statistics: *10th Percentile*, *90th Percentile* (∼RMS), *Variance*, and *Skewness* were the most impactful contributors to model predictions. These features not only ranked highly but also showed consistent directional effects: higher values tended to push predictions toward the basal class, while lower values supported luminal classification. Among texture features, *GLCM Cluster Tendency*, *GLRLM Long Run Low Gray Level Emphasis*, *NGTDM Strength*, and *NGTDM Busyness* showed moderate influence, indicating that while most texture features underperformed, a select few retained predictive value.

Overall, these findings validate that a curated set of radiomic features, primarily first-order metrics, can reliably distinguish tumour phenotypes in photoacoustic imaging. Their stability across confounders and alignment between statistical and model-based importance rankings supports their potential as robust, biologically meaningful imaging biomarkers.

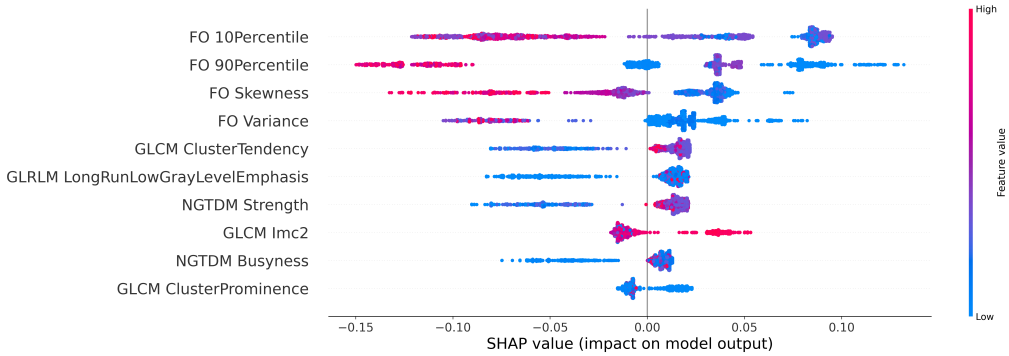

Figure 2: SHAP beeswarm plot for the Random Forest classifier, highlighting the top 10 most influential radiomic features. Blue and red indicate low and high feature values, respectively.

3 Part B: LIDC-IRDI Extension

To explore the broader clinical applicability of radiomic pipelines, we extended the analysis to the LIDC-IDRI dataset of annotated lung CT scans. This tested whether the methodologies developed for the controlled

photoacoustic dataset could generalise to malignancy detection in a more complex and heterogeneous clinical setting. From 107 patients, we segmented 284 3D nodules and extracted 107 radiomic features, including first-order, shape, and texture descriptors. We further evaluated the impact of class balancing via SMOTE and compared performance across Recursive Feature Elimination (RFE) and Boruta feature selection strategies.

ML Classification Performance We benchmarked four classifiers: Random Forest, XGBoost, Logistic Regression, and SVM, using features selected via RFE (27 features), both with and without SMOTE class balancing (Fig.3). Although Boruta based models were also evaluated, they consistently underperformed relative to their RFE counterparts across most metrics, and are therefore omitted from Fig.3 for clarity. In the imbalanced dataset (229 malignant vs. 55 benign), models trained without SMOTE generally achieved higher overall accuracy but exhibited poor recall for benign tumours. For example, Random Forest reached 79% accuracy but only 30% recall for Class 0. Applying SMOTE substantially improved sensitivity, with SVM recall increasing from 48% to 59%, albeit with a trade-off in Class 1 precision.

The best class balance was achieved by the SVM model with RFE features and SMOTE, yielding F1 scores of 0.420 (benign) and 0.783 (malignant). While performance remains modest, these results are reasonable given the limited dataset size and imbalance.

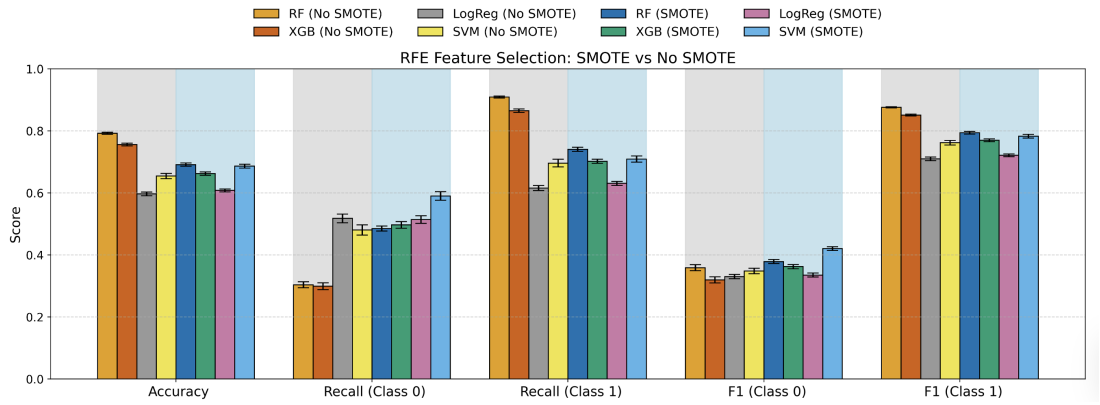

Figure 3: Performance comparison of four classifiers across the RFE feature set.

SHAP Analysis SHAP analysis provided further insight into feature influence. *GLSZM Gray Level Non-Uniformity* consistently emerged as one of the most impactful features. Higher values (indicative of heterogeneity) aligned with malignancy predictions. This consistency held across resampling strategies and model seeds (see Fig.5), reinforcing its potential as a stable biomarker. Additional features such as *GLCM Joint Energy*, *NGTDM Coarseness*, and *GLCM Cluster Shade* also contributed meaningfully.

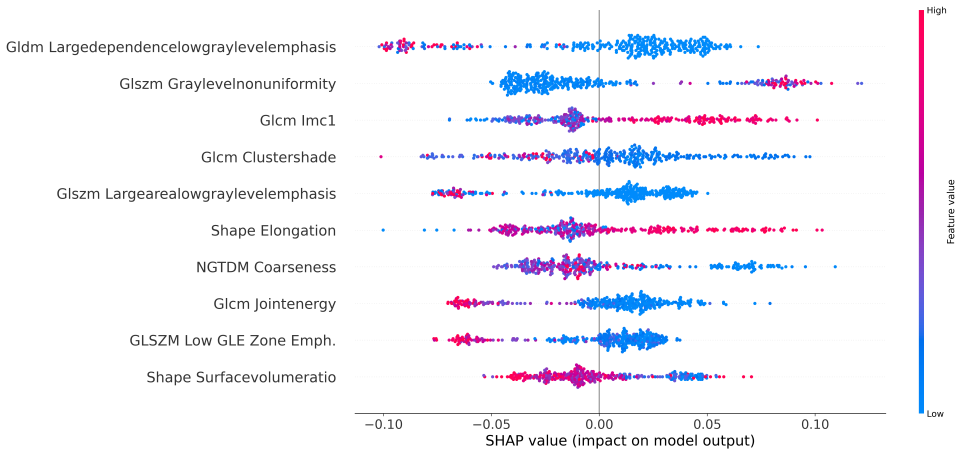

Figure 4: SHAP beeswarm plot for the Random Forest model trained on the SMOTE-balanced dataset.

These results show that, even in complex clinical CT data, a well-tuned pipeline, combining class balancing and interpretable feature selection, can still identify meaningful radiomic markers. This highlights the potential to generalise radiomic methods beyond controlled experimental settings.

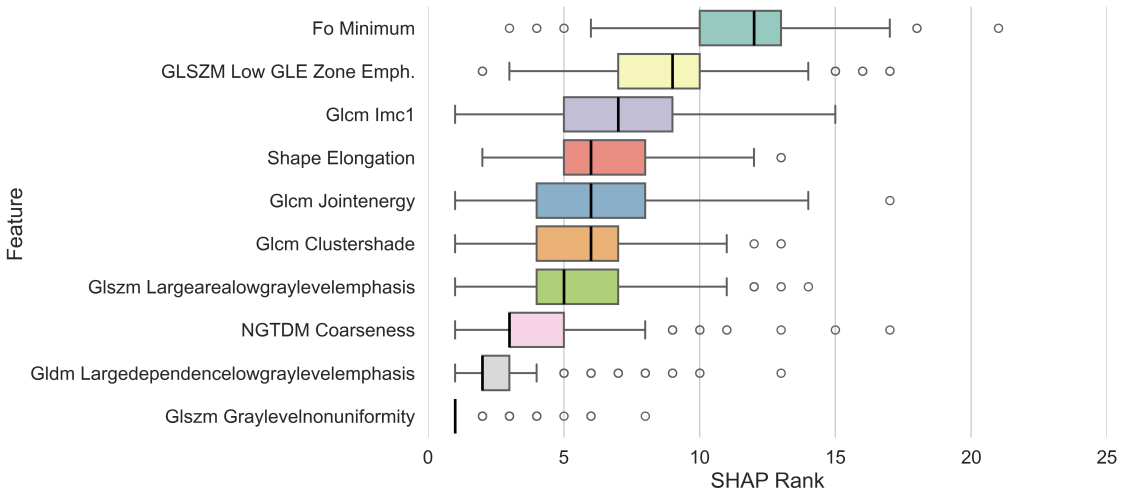

Figure 5: SHAP feature rank distribution across 1000 training seeds for the SMOTE-balanced dataset. Features are ordered by median rank.

4 Conclusion

This study identified modality-specific radiomic features with strong predictive value. In photoacoustic imaging, first-order metrics like *Skewness* and *Kurtosis* reflected tumour biology with minimal confounding. In the LIDC extension, *GLSZM Gray Level Non-Uniformity* consistently ranked as the most informative feature. However, with only 280 tumours analysed, these findings should be seen as exploratory. Robust clinical translation will require far larger datasets and standardised protocols to validate the reliability of radiomic biomarkers.